A HARMONISED POINT-EXTRACTION AND COMPARISON TOOL FOR HETEROGENEOUS PALEOCLIMATE DATASETS

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ABSTRACT:

Paleoclimate simulations are central to understanding how the Earth system behaved before instrumental records existed, but the datasets that hold these simulations are published in formats that do not line up with one another. They use different time origins and units, different spatial grids and longitude conventions, different temperature variables, and different temporal and spatial resolutions. This makes even a simple side-by-side comparison awkward. This project provides a reproducible Python pipeline that downloads, preprocesses, and compares the near-surface temperature archives of four widely used datasets: Beyer2020/2021, PalMod2, TraCE-21k, and CHELSA-TraCE21k-centennial. The pipeline brings all four onto a common kiloyears-before-present (ka BP) time axis, resolves the differing spatial grids and longitude conventions, and converts every temperature field to degrees Celsius. From the harmonised data the tool extracts a temperature time series at any user-defined point, and it compares each pair of datasets globally by sampling every model on the shared coarse T31 grid and mapping the mean temperature difference per cell. At the example location the datasets agree on the broad shape of the deglaciation, a cold interval around 20 ka BP followed by warming into the Holocene, but they disagree on absolute temperature by several degrees. The global difference maps show that the disagreement is not random: it is largest at high northern latitudes and over high terrain, and smallest in the mid-latitude lowlands and the tropics. This report documents the four datasets, the acquisition method, the preprocessing and harmonisation steps, the point extraction, and the global comparison. The full pipeline is organised as a set of Jupyter notebooks (Giersch, 2026).

1. INTRODUCTION

1.1 Why comparing paleoclimate models matters

Paleoclimate models reconstruct the state of the climate for periods that predate direct measurement. They are used to study how the climate responds to forcings such as solar output, volcanic eruptions, orbital cycles, and atmospheric CO_2, and they provide the natural variability baseline against which recent, human-driven change is judged. The same simulations help constrain climate sensitivity, the temperature change expected per doubling of CO_2, and past warm intervals are sometimes used as rough analogues for near-future projections. Methods developed for terrestrial paleoclimate also carry over to the climate evolution of other planetary bodies.

Because there are many such models, and because they are built from different physics, forcings, and resolutions, it is useful to compare them directly. A comparison shows where the models agree, which is a sign that a feature is robust, and where they diverge, which flags the regions and quantities that are still poorly constrained. Outliers and systematic offsets between models are exactly the kind of signal that motivates a closer look at a particular model or region.

1.2 Aim of this project

The aim of this project is narrow and practical. It is to take four diverse and individually incompatible paleoclimate datasets and make them directly comparable, then to build a small tool on top of the harmonised data that does two things. First, it extracts a temperature time series, air or soil, at a point chosen by the user and plots all available datasets on one axis. Second, it compares the datasets pairwise across the whole globe, sampling each one on the resolution of the coarse T31 grid shared by two of the models, and presents the mean temperature differences as maps. The project does not attempt to judge which model is correct. It documents the differences and shows where they concentrate.

2. THE DATA

The comparison uses four datasets, summarised in Table 1. They were chosen because they overlap in time over the last deglaciation while differing in almost every technical respect, which makes them a good stress test for a harmonisation pipeline. Two of them, PalMod2 and TraCE-21k, also provide soil temperature, so the tool handles that variable as well.

2.1 TraCE-21k

TraCE-21k is a transient simulation of the climate of the last 21,000 years (Otto-Bliesner and Rosenbloom, 2021). It was produced with the Community Climate System Model version 3 (CCSM3) as a fully coupled, non-accelerated atmosphere, ocean, sea-ice, and land-surface run, so the simulation integrates forward in real model years from the Last Glacial Maximum to the recent past rather than stepping between equilibrium snapshots. The version used here is the main land decadal-average product, which stores ten-year means on the coarse T31_gx3 grid. Two variables are taken from it: `TSA`, the 2 m near-surface air temperature from the CLM2 land component, and `TSOI`, the soil temperature by layer. The two files together are roughly 450 MB. Time runs from about 22 ka BP to the present in 2204 decadal steps.

2.2 CHELSA-TraCE21k

CHELSA-TraCE21k-centennial is a high-resolution downscaling of TraCE-21k (Karger et al., 2023, Karger et al., 2020). It

Dataset	Variables	Time coverage	Time step	Spatial resolution	Format
Beyer2020 (v4)	air temp	120–0 ka BP	1–2 kyr	0.5°	NetCDF
PalMod2	air temp, soil temp	25–0 ka BP	annual	~3.75° (T31)	NetCDF
TraCE-21k	air temp, soil temp	22–0 ka BP	decadal	~3.75° (T31)	NetCDF
CHELSA-TraCE21k-centennial	air temp	22–0 ka BP	100 yr	~1 km (0.01°)	GeoTIFF → NetCDF

Table 1. Overview of the four paleoclimate datasets used in the comparison.

was generated with the CHELSA V1.2 algorithm, which reconstructs a paleo-orography for each time step and then applies lapse-rate and orographic-precipitation downscaling, producing monthly climatologies at roughly 1 km resolution (0.01°, a global grid of 43200 × 20880 cells) in 100-year steps since the Last Glacial Maximum. The dataset is distributed as monthly GeoTIFFs of `tasmin` and `tasmax`, with temperature stored as Kelvin multiplied by ten. For this project the two variables are averaged into a mean air temperature, $\mathrm{tasmean} = (\mathrm{tasmin} + \mathrm{tasmax})/2$. Because the dataset inherits its large-scale signal from TraCE-21k but adds a separate bias-correction and downscaling step, it is a useful test of whether the fine-scale product still resembles its coarse parent when both are sampled on the same grid. Time covers roughly 22 ka BP to the present in 221 centennial steps.

2.3 Beyer2020 / 2021

The Beyer dataset is a high-resolution reconstruction of terrestrial climate, bioclimate, and vegetation for the last 120,000 years, produced by Beyer, Krapp, and Manica at the University of Cambridge (Beyer et al., 2020, Beyer et al., 2021). It fills the gap between paleoclimate products that have good temporal resolution but coarse grids and those that have fine grids but only cover a handful of snapshots. The reconstruction combines medium-resolution HadCM3 transient simulations with high-resolution HadAM3H snapshots and present-day observations, using a two-stage variant of the Delta method to downscale and bias-correct the fields to a 0.5° equirectangular grid (about 55 km at the equator). Gridded fields are provided at 2000-year steps between 120 and 22 ka BP and at 1000-year steps from 22 ka BP to the pre-industrial present, giving 72 snapshots in total. The version used here is v4 on figshare (Beyer, 2020a, Beyer, 2020b), which adds global monthly net primary productivity relative to v3 and is documented in the 2021 addendum. The single bundled NetCDF stores monthly mean temperature in degrees Celsius, which makes it the only one of the four already in the target unit. Of the bundled variables, only `temperature` (air) is used.

2.4 PalMod2

PalMod2 is a transient simulation of the last deglaciation run with the Max Planck Institute Earth System Model version 1.2 in its coarse-resolution configuration, MPI-ESM1-2-CR (Mikolajewicz et al., 2023). The atmosphere is the ECHAM6.3 spectral model at T31 (96 by 48 longitude and latitude cells, 31 vertical levels), coupled to the MPIOM ocean, the JSBACH land model, and a sea-ice model, with time-evolving ice sheets, topography, and land-sea mask prescribed from GLAC-1D reconstructions. The specific product is the r1i1p1f1 realisation of the transient deglaciation experiment with prescribed GLAC-1D ice sheets. Two monthly variables are used: `tas`, near-surface air temperature (dataset id `PMMXMCRTDGr111Amtasgn30201`), and `tsl`, soil temperature by layer (`PMMXMCRTDGr111Lmtslgn30201`). The data are served from the World Data Center for Climate (WDCC) at DKRZ and downloaded with the `jblob` client after registration (for Climate and Klimarechenzentrum, n.d.). Both variables are stored in Kelvin. Time runs from about 25 ka BP to the present at monthly resolution, which the pipeline reduces to 25,000 annual means.

3. DATA ACQUISITION

All four datasets are fetched by a single notebook, `data_downloader.ipynb`, so that the whole archive can be reconstructed from scratch on any machine. Each source needs a slightly different access method, and the notebook keeps them in one place.

TraCE-21k is downloaded from NCAR by reading a list of file URLs and keeping only the `TSA` and `TSOI` variables. Beyer2020 is fetched through the figshare API: the notebook queries version 4 of article `12293345` for its file list and downloads the bundled NetCDF. CHELSA is the largest source by far. Its file URLs are read from an EnviDat path list, filtered to `tasmin` and `tasmax`, and downloaded in parallel with a 32-worker thread pool, since the full set of monthly GeoTIFFs is large and download-bound. PalMod2 is the most involved: it requires a WDCC account, and the `jblob` command-line client is driven through `pexpect` to handle the interactive login, after which the `tas` and `tsl` archives are pulled to a temporary path and moved into the data layout. The total download is dominated by CHELSA and reaches roughly 1 TB.

4. PRE-PROCESSING

TraCE-21k and Beyer2020 arrive as analysis-ready NetCDF files and need no preprocessing. The other two do, and the work is done in `data_processer.ipynb`.

4.1 PalMod2

PalMod2 `tas` and `tsl` are delivered as many monthly NetCDF files inside tar archives. For each file the pipeline computes an annual mean with `cdo yearmean`, collapsing twelve monthly values into one yearly value, and then merges all of the yearly files along the time axis with `cdo mergetime` into a single output per variable. The merged air-temperature file has 25,000 annual time steps. The CDO calls are run from inside the extraction directory using relative file names, which avoids a path-handling problem caused by spaces in the original data directory.

4.2 CHELSA-TraCE21k

CHELSA needs the most work. The monthly GeoTIFFs are first grouped by year using a regular expression on the filenames, which encode the month and a CHELSA time index; the index is mapped to a calendar year with the anchor that index 20 corresponds to 1990 CE and each step is 100 years. For each year the twelve monthly tiles are stacked and averaged, rescaled from Kelvin times ten to Kelvin with a factor of 0.1, and tagged with a July-1 timestamp. The yearly slices are concatenated along time, given metadata, and written as one gzip-compressed NetCDF per variable. A separate step then computes `tasmean` as the mean of `tasmin` and `tasmax`.

This stage is the bottleneck of the whole pipeline. At full 1 km resolution the intermediate data run to about 1 TB of disk and several days of computation on a modern laptop. CDO cannot be used here because the arrays do not fit in memory, so the processing is built on `xarray` and `rioxarray` with Dask, which keeps the computation lazy and streams it chunk by chunk to disk. The chunk size, 4500×2250 pixels, sets the peak memory use, and the number of Dask workers is capped to keep that bounded (Giersch, 2026).

5. HARMONISATION

To compare the datasets they have to share a common variable, time format, spatial reference, and unit. The harmonisation is implemented in the comparison notebook (Giersch, 2026) and is applied on the fly when data are read, so the large source files are never rewritten.

5.1 Variables and units

Each dataset is described by a small registry that records, for every temperature field, its file path, the variable name inside the file, and its native unit. The air-temperature registry covers all four datasets; the soil-temperature registry covers only PalMod2 and TraCE-21k. Three of the four datasets store temperature in Kelvin, while Beyer2020 uses degrees Celsius, so the reader subtracts 273.15 from the Kelvin sources and leaves Beyer unchanged. All output is in degrees Celsius. Where a dataset carries a seasonal cycle, the twelve monthly values are averaged to an annual mean before anything else, and where a soil variable has several depth layers, the topmost layer is taken.

5.2 Time

The time axis is the hardest part, because every file uses a different convention, summarised in Table 2. A single helper, `time_to_ka_bp`, normalises all four to kiloyears before present with the past positive and the present at zero, so that everything downstream can treat them uniformly. Beyer stores years since present with the past as negative values, which become ka BP by dividing by minus 1000. TraCE stores values already labelled ka BP but written as negative kiloyears, so they are simply negated. PalMod2 and CHELSA store days since a calendar reference date that runs forward toward the present; for these the absolute reference is irrelevant, and the conversion anchors on the most recent sample and turns the elapsed days into kiloyears. That last step is accurate to about half a kiloyear, which is invisible on the time axis used here.

The four datasets also differ greatly in temporal resolution and coverage. Beyer reaches the furthest back, to 120 ka BP, but in coarse steps of 1000 to 2000 years (72 samples). PalMod2 is annual (25,000 samples over 25 ka), TraCE is decadal (2204 samples over 22 ka), and CHELSA is centennial (221 samples over 22 ka). Any pairwise comparison is therefore restricted to the overlapping time window of the two datasets involved.

5.3 Space

The spatial grids are equally varied. Beyer is a 0.5° grid on the -180 to 180 longitude convention. PalMod2 and TraCE share the coarse T31 grid, 96 by 48 cells at about 3.71° latitude by 3.75° longitude, on the 0 to 360 convention. CHELSA is a 0.01° grid on the -180 to 180 convention. Two issues follow. First, the longitude convention has to be reconciled before any query, so the reader adds 360 to a negative requested longitude whenever the file uses the 0 to 360 range. Second, the cell that a given point falls into is found by nearest-neighbour selection (`xarray.sel(method="nearest")`), which means the same requested point returns cells of very different size depending on the dataset, and not always the same geographic centre. The T31 grid also has cells whose latitude width is not exactly constant, a feature of the underlying Gaussian grid.

6. POINT EXTRACTION

Point extraction opens one file, selects the grid cell nearest to the requested latitude and longitude, collapses any seasonal cycle to an annual mean, takes the topmost layer for soil temperature, converts Kelvin to Celsius if needed, and returns the resulting one-dimensional series on the ka BP axis. Because the selection is nearest-neighbour rather than interpolation, the value returned is the native cell value, with no smoothing across cells. This is a deliberate choice: it keeps each dataset's own representation intact, at the cost that a coarse dataset is represented at a point by a single large cell while a fine dataset is represented by a small one. The same routine drives both the time-series plot and the global difference maps, so the two cannot drift apart.

7. RESULTS

7.1 An example point

Figure 1 overlays the four air-temperature series at an example location in northern Germany, 53.26°N, 11.67°E. The legend reports the actual cell each dataset returned, and the cells differ: Beyer returns 53.25°N, 11.75°E, the coarse T31 grid of PalMod2 and TraCE snaps to 53.81°N, 11.25°E, and CHELSA returns 53.25°N, 11.67°E. This is the resolution effect made visible.

Two things stand out. The datasets agree on the shape of the deglaciation: a cold interval around 20 to 22 ka BP, near or below minus 10°C at this site for the deglacial models, followed by warming into the Holocene. They disagree on the absolute Holocene temperature by several degrees. Beyer, PalMod2, and CHELSA settle near 8 to 9°C, while TraCE settles around 3°C, making it the cold outlier here. The series also differ in smoothness, which is a direct consequence of their temporal resolution: Beyer is smooth because it has only one value per one to two thousand years, whereas the annual, decadal, and centennial series carry visible year-to-year and century-to-century variability.

The companion summary, Figure 2, makes the coverage and resolution differences explicit. The left panel shows each dataset's temporal extent and median step on a symmetric-log axis, and the right panel draws, at true size, the grid cell each dataset returns near the requested point. The contrast between the small Beyer and CHELSA cells and the large shared T31 cell

Dataset	Raw time units	Raw range	Conversion to ka BP
Beyer2020	`years since present` (past negative)	$[-120000,\ 0]$	$-$vals/1000
PalMod2	`days since 1-1-1` (forward to present)	$[181,\ 9\,130\,878]$	(max $-$vals)/365250
TraCE-21k	`ka BP` (stored negative)	$[-22,\ 0.03]$	$-$vals
CHELSA	`days since -20010-07-01` (forward)	$[0,\ 8\,035\,335]$	(max $-$vals)/365250

Table 2. Native time conventions of the four datasets and the conversion applied by `time_to_ka_bp` to bring them onto a common ka BP axis (past positive, present zero).

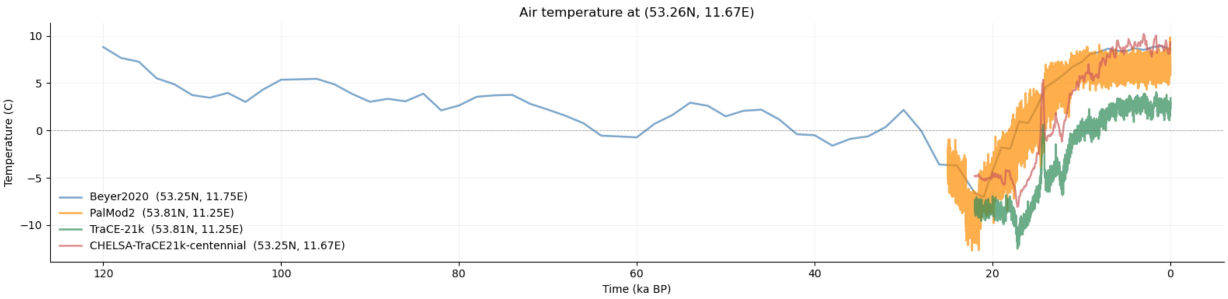

Figure 1. Air-temperature time series at 53.26°N, 11.67°E for all four datasets on a common ka BP axis. The datasets agree on the deglacial warming but differ in absolute temperature, in temporal smoothness, and in the grid cell each one samples (see legend).

of PalMod2 and TraCE is the clearest single illustration of why a naive comparison is misleading.

PalMod2 and TraCE are worth singling out, because they sit on the identical T31 cell at this point (53.81°N, 11.25°E in both legends). They track each other closely through the deglaciation but separate after about 10 ka BP, with PalMod2 holding several degrees warmer than TraCE through the Holocene (Figure 1). Since the two share the same grid cell, this gap is a genuine model difference and not an artefact of resolution or location.

7.2 Global pairwise differences

A single point cannot show whether these offsets are local or systematic, so the tool repeats the comparison everywhere. For each pair of datasets it samples both on the cell centres of the shared T31 grid, 96 by 48 = 4608 cells, averages each over the time window the two share, and maps the per-cell difference. The maps use a diverging green-to-purple scale on which green means the first-named dataset is warmer, purple means it is colder, and white means the two agree. Sampling everything on the coarse T31 grid means that for the fine datasets, Beyer and CHELSA, each large cell is represented by a single nearest-neighbour point rather than an area average, which is a limitation discussed below.

The six air-temperature pairings are shown in Figures 3 to 8. A consistent pattern runs through all of them. The differences are small over the mid-latitude lowlands and the tropics, and they grow large at high northern latitudes, over the Arctic, Scandinavia, Siberia, and Greenland, and over high terrain such as the Tibetan Plateau and the Andes. In other words, the datasets agree most where the climate is mild and the terrain is simple, and disagree most where it is cold or mountainous, which are also the regions where ice-sheet history, elevation, and the choice of near-surface variable matter most.

The individual pairings refine this picture. Beyer minus PalMod2 (Figure 3) shows Beyer colder over the Arctic and northern Eurasia and warmer over high terrain. Beyer minus TraCE (Figure 4) is green across almost the whole northern hemisphere, meaning Beyer is warmer than TraCE nearly everywhere at mid to high latitudes, which is the global counterpart of the cold TraCE outlier seen at the example point.

Beyer minus CHELSA (Figure 5) is the opposite sign at high latitudes: Beyer is colder than CHELSA (purple) over the Arctic and northern Europe. This is notable because CHELSA is downscaled from TraCE, yet sampled on the T31 grid it behaves quite differently from its parent. The TraCE minus CHELSA map (Figure 8) confirms this directly: TraCE is colder than CHELSA (purple) over most of the northern high latitudes. The downscaling and bias-correction steps that produce CHELSA therefore do not simply reproduce the coarse TraCE field; at the cell centres compared here CHELSA is systematically warmer in the north.

Among the deglacial models, PalMod2 minus TraCE (Figure 6) shows PalMod2 warmer than TraCE (green) across the high northern latitudes, with smaller patches of the opposite sign over central Asia and the Tibetan Plateau. This is the spatial generalisation of the Holocene gap seen at the example point in Figure 1. PalMod2 minus CHELSA (Figure 7) is more mixed and speckled, without a single dominant sign, which fits CHELSA being warmer than TraCE while PalMod2 is also warmer than TraCE, so the two warm models land closer to each other.

The same comparison for soil temperature, available only for PalMod2 and TraCE, is shown in Figure 9. The spatial pattern mirrors the air-temperature pairing of the same two models: PalMod2 has warmer soils than TraCE across the high northern latitudes and cooler soils over central Asia and the Tibetan Plateau. That the soil and air differences share a pattern is a basic consistency check, since the two variables are physically linked.

Taken together, the maps support the conclusion drawn from the single point. The datasets agree on the large-scale story of the deglaciation but differ by several degrees in absolute temperature, and those differences are concentrated at high northern latitudes and over high terrain rather than spread evenly across the globe.

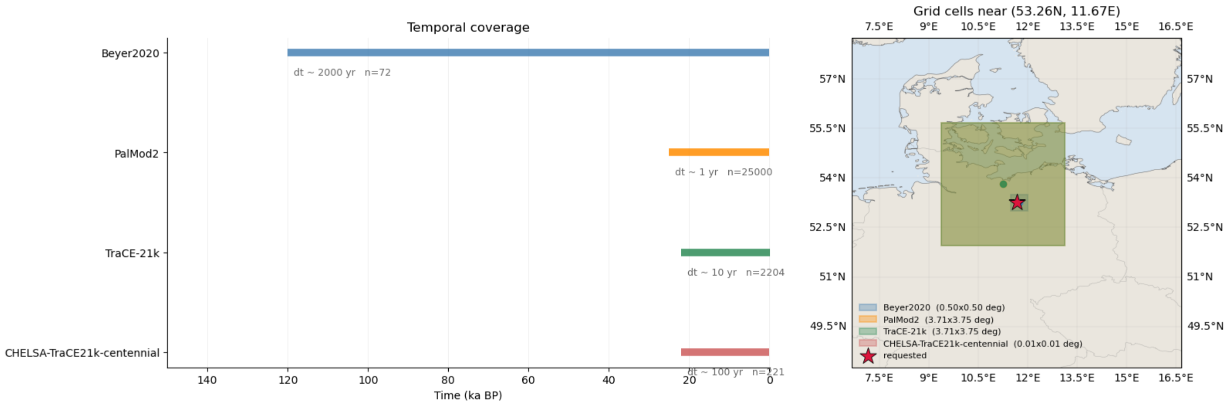

Figure 2. Left: temporal coverage and median time step of each dataset (note the very different sample counts n). Right: the grid cell each dataset returns near 53.26°N, 11.67°E, drawn at its true size in degrees; the red star is the requested point. The T31 cell of PalMod2 and TraCE dwarfs the Beyer and CHELSA cells.

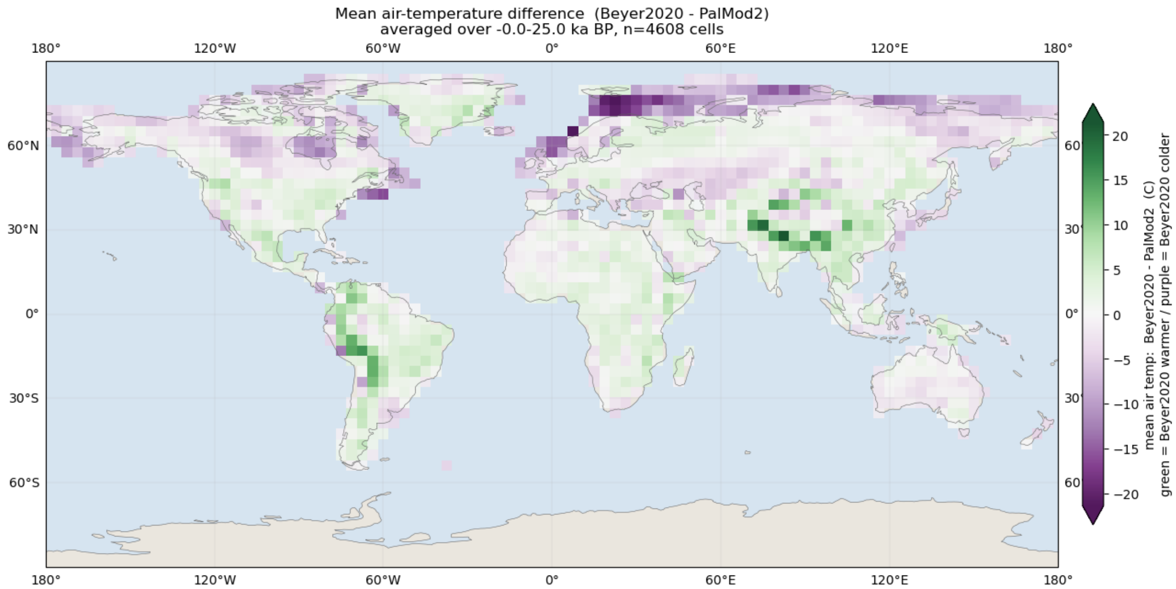

Figure 3. Mean air-temperature difference Beyer2020 minus PalMod2, averaged over the shared 0–25 ka BP window. Beyer is colder (purple) than PalMod2 across the high northern latitudes and warmer (green) over the Tibetan Plateau, central Asia, and parts of South America.

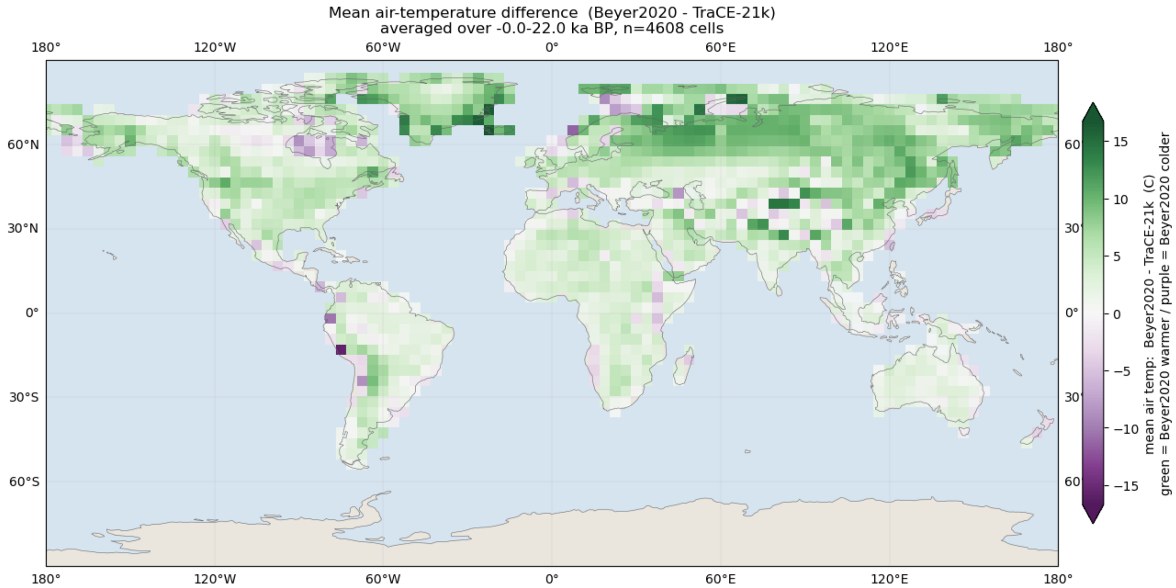

Figure 4. Mean air-temperature difference Beyer2020 minus TraCE-21k over 0–22 ka BP. Beyer is warmer than TraCE (broad green) across the northern hemisphere mid and high latitudes, consistent with TraCE being the cold outlier.

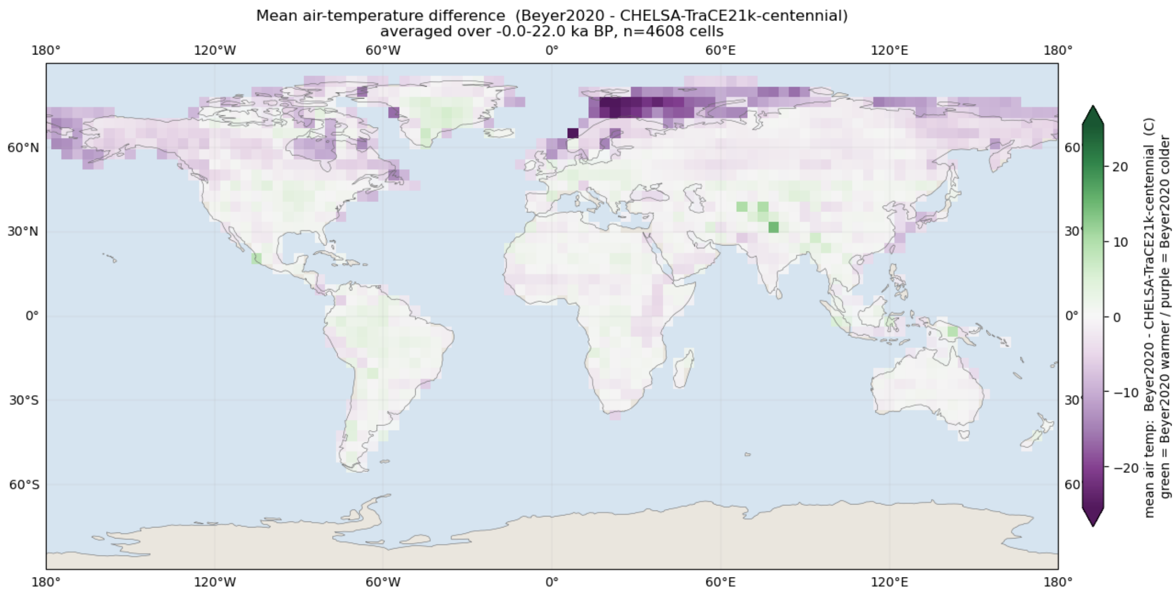

Figure 5. Mean air-temperature difference Beyer2020 minus CHELSA-TraCE21k over 0–22 ka BP. Beyer is colder than CHELSA (purple) over the high northern latitudes, the opposite sign to the Beyer minus TraCE map.

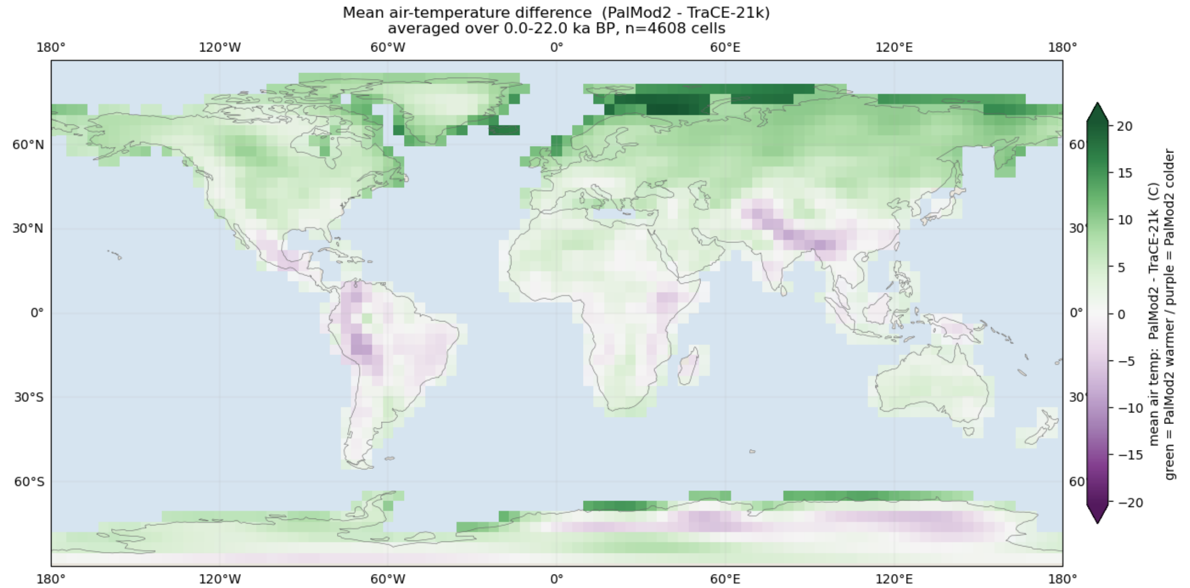

Figure 6. Mean air-temperature difference PalMod2 minus TraCE-21k over 0–22 ka BP. PalMod2 is warmer (green) across the high northern latitudes, with localised colder patches over central Asia and the Tibetan Plateau.

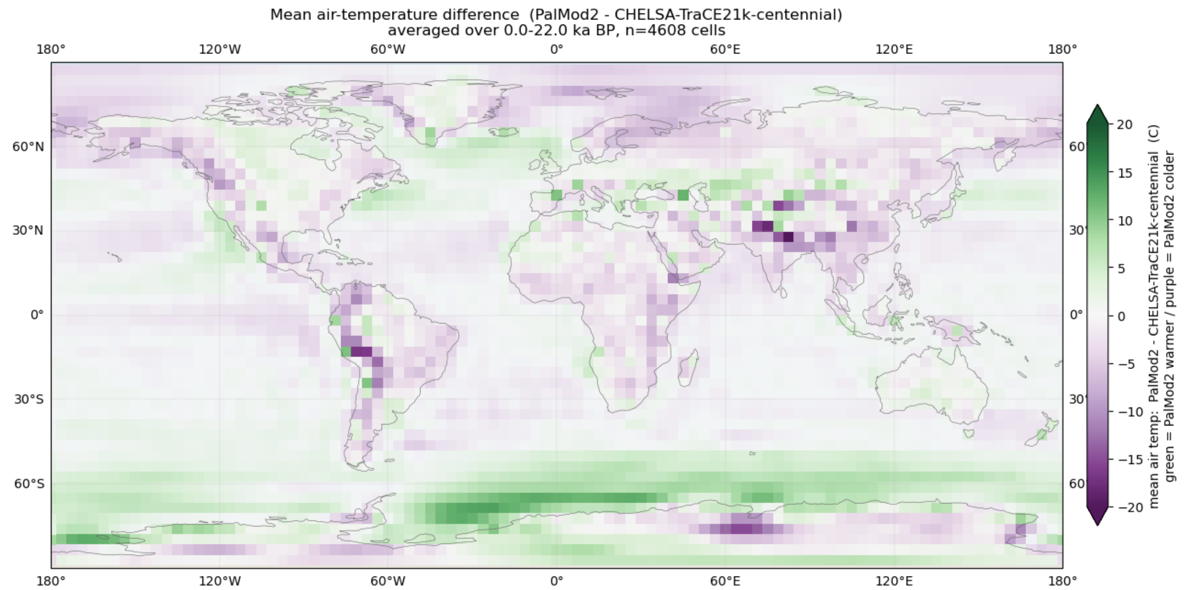

Figure 7. Mean air-temperature difference PalMod2 minus CHELSA-TraCE21k over 0–22 ka BP. The pattern is mixed, without a single dominant sign, reflecting that both PalMod2 and CHELSA are warmer than TraCE.

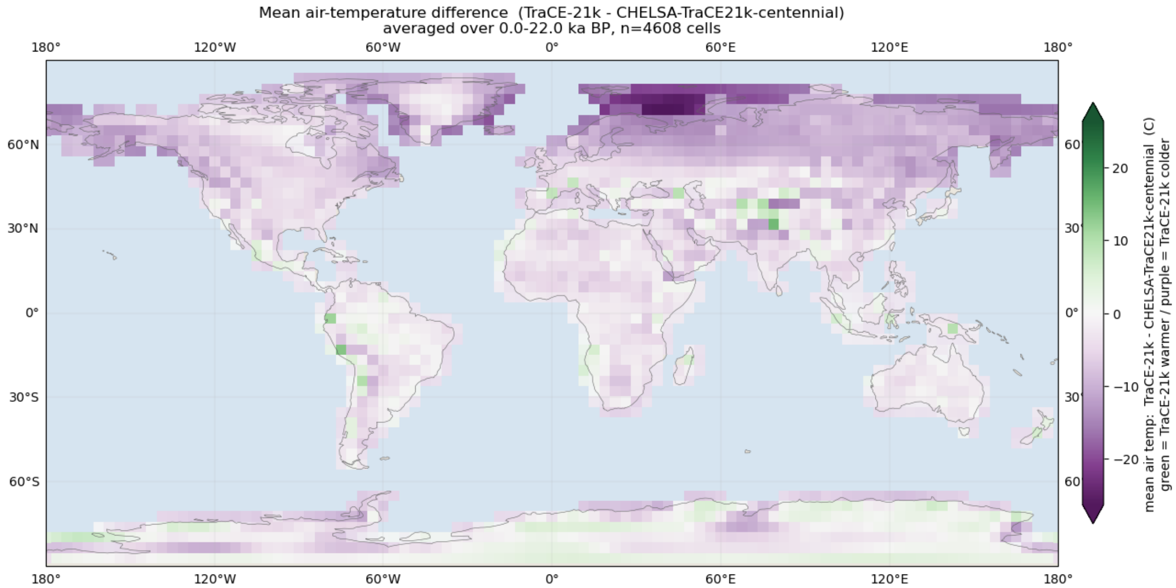

Figure 8. Mean air-temperature difference TraCE-21k minus CHELSA-TraCE21k over 0–22 ka BP. TraCE is colder than its own downscaled product CHELSA (purple) across most of the northern high latitudes.

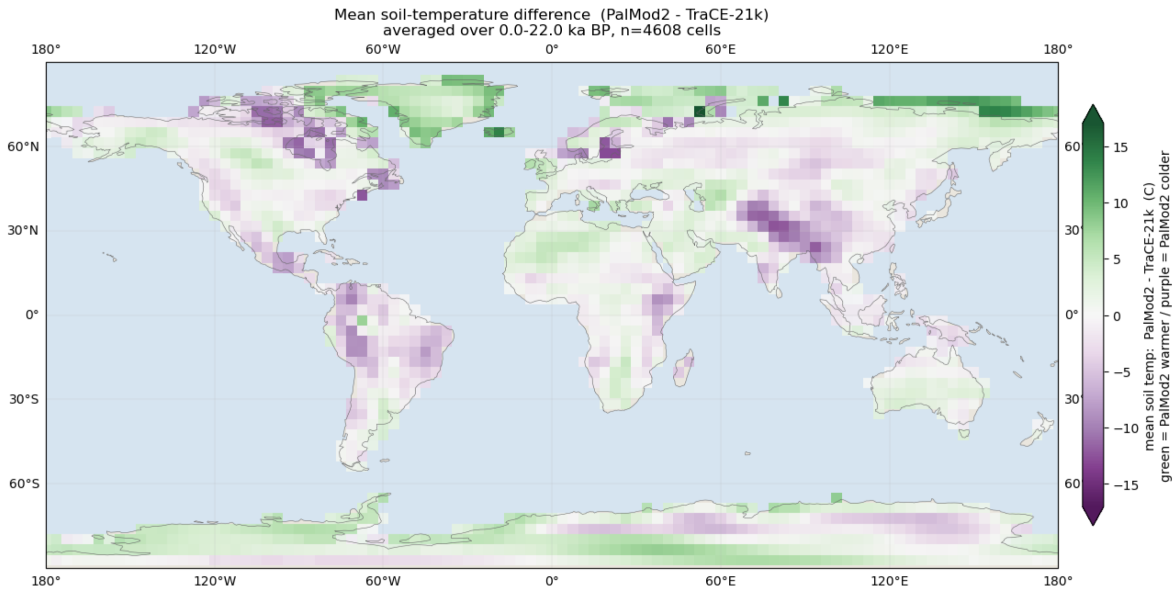

Figure 9. Mean soil-temperature difference PalMod2 minus TraCE-21k over 0–22 ka BP. The spatial pattern follows the air-temperature pairing of the same two models, a consistency check between the two physically linked variables.

8. DISCUSSION AND LIMITATIONS

A few caveats should temper the maps. The comparison samples every dataset by nearest-neighbour at the centres of the coarse T31 cells, so the fine grids of Beyer and CHELSA are reduced to one point per large cell instead of being averaged over the cell area. In rugged terrain that single point can sit at an elevation far from the cell mean, which inflates the apparent difference exactly in the regions, mountains and ice margins, where the disagreement is already largest. An area-weighted aggregation of the fine grids onto the T31 cells would be the more defensible comparison and would likely shrink some of the extreme values. The time anchoring for the two day-count datasets is also approximate to about half a kiloyear, and the pairwise windows differ because the datasets cover different spans, so the averages behind each map are not all taken over the same period. None of these undermines the main finding, that the disagreement is structured and concentrated, but they do caution against reading the precise magnitude in any single cell too literally.

9. FUTURE WORK

The current tool compares one point at a time and inspects the global difference maps by eye. The natural next step is to make the analysis quantitative. The difference maps could be paired with a per-cell correlation analysis through time, testing each cell's pair of series for a normal distribution and then applying a Pearson or a rank-based Spearman correlation as appropriate, which would separate cells where the datasets merely differ by an offset from cells where they disagree about the shape of the change. Correlating across space would give a time series of agreement, and correlating across time would give a map of agreement per cell, which read together with the difference maps would be a stronger basis for discussion. The differences could also be related to explanatory variables such as latitude, elevation, land versus ocean, and climate zone, to test directly the impression that the disagreement tracks cold and mountainous regions. Finally, the point extractor could be looped automatically over many sites rather than run one location at a time.

10. CONCLUSION

The four datasets are genuinely hard to compare, because they differ in time origin, unit, grid, longitude convention, temperature variable, and resolution. The pipeline described here removes those obstacles in a reproducible way and reduces every dataset to a common ka BP axis, a common unit, and a common grid for comparison, then exposes both a point tool and a global difference map for each pair. Used on these four datasets, the tool shows agreement on the broad deglacial trend and disagreement of several degrees in absolute temperature, concentrated at high northern latitudes and over high terrain. A comparison of this kind cannot say which model is right, but it can point to where they diverge most, and those regions are the obvious candidates for closer scrutiny and, in turn, for improving the models.

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